

10.18 Consider the data file *mroz* on working wives. Use the 428 observations on married women who participate in the labor force. In this exercise, we examine the effectiveness of a parent's college education as an instrumental variable.

- Create two new variables. *MOTHERCOLL* is a dummy variable equaling one if *MOTHEREDUC* > 12, zero otherwise. Similarly, *FATHERCOLL* equals one if *FATHEREDUC* > 12 and zero otherwise. What percentage of parents have some college education in this sample?
- Find the correlations between *EDUC*, *MOTHERCOLL*, and *FATHERCOLL*. Are the magnitudes of these correlations important? Can you make a logical argument why *MOTHERCOLL* and *FATHERCOLL* might be better instruments than *MOTHEREDUC* and *FATHEREDUC*?
- Estimate the wage equation in Example 10.5 using *MOTHERCOLL* as the instrumental variable. What is the 95% interval estimate for the coefficient of *EDUC*?
- For the problem in part (c), estimate the first-stage equation. What is the value of the *F*-test statistic for the hypothesis that *MOTHERCOLL* has no effect on *EDUC*? Is *MOTHERCOLL* a strong instrument?
- Estimate the wage equation in Example 10.5 using *MOTHERCOLL* and *FATHERCOLL* as the instrumental variables. What is the 95% interval estimate for the coefficient of *EDUC*? Is it narrower or wider than the one in part (c)?
- For the problem in part (e), estimate the first-stage equation. Test the joint significance of *MOTHERCOLL* and *FATHERCOLL*. Do these instruments seem adequately strong?
- For the IV estimation in part (e), test the validity of the surplus instrument. What do you conclude?

a.

percentage that mother has college education : 12.1495 %

percentage that father has college education : 11.6822 %

b.

correlation between *EDUC* and *MOTHERCOLL* : 0.3595

correlation between *EDUC* and *FATHERCOLL* : 0.3985

correlation between *MOTHERCOLL* and *FATHERCOLL* : 0.3546

c.

95 % CI for *EDUC* $\Rightarrow [-0.001219763, 0.1532359]$

d.

$F = 63.563$

$p\text{-value} = 1.455 \times 10^{-14}$

$\Rightarrow$ reject H_0

e.

95 % CI for *EDUC* $\Rightarrow [0.02751845, 0.1481969]$

f.

$F = 56.963$

$p\text{-value} = 2.2 \times 10^{-16}$

$\Rightarrow$ reject H_0

10.18

a.

```
> # Calculate percentage
> mean(mroz_working$mothercoll)
[1] 0.1214953
> mean(mroz_working$fathercoll)
[1] 0.1168224
```

- The percentage of mothers with some college education in the sample is **12.1495%**.
- The percentage of fathers with some college education in the sample is **11.6822%**.

b.

```
> cor(mroz_working[c("educ", "mothercoll", "fathercoll")])
          educ mothercoll fathercoll
educ      1.0000000  0.3594705  0.3984962
mothercoll 0.3594705  1.0000000  0.3545709
fathercoll 0.3984962  0.3545709  1.0000000
```

- The correlation between EDUC and MOTHERCOLL is **0.3595**.
- The correlation between EDUC and FATHERCOLL is **0.3985**.
- The correlation between MOTHERCOLL and FATHERCOLL is **0.3546**.
- **The magnitudes of these correlations are important.** They indicate a moderately positive linear relationship, suggesting that parents' college education levels are relevant to the wife's education level. This **satisfies the "relevance" condition for a valid instrument variable.** Additionally, MOTHERCOLL and FATHERCOLL might be better instruments than MOTHEREDUC and FATHEREDUC because attaining a college degree often represents a significant threshold in family socioeconomic status and resources, which can more distinctly influence the offspring's education opportunities.

c.

- Using MOTHERCOLL as the single instrumental variable, the 95% confidence interval for the coefficient of EDUC is **[-0.001219763, 0.1532557]**.

```
> # Calculate the 95% CI for educ
> confint(mroz1_iv, "educ", level = 0.95)
          2.5 %    97.5 %
educ -0.001219763 0.1532557
```

d.

- F-statistic = **63.563**
- P-value = **1.455e-14**
- Since the **p-value is extremely small (close to 0)**, we **reject the null hypothesis**. Moreover, according to the rule of thumb in econometrics, an F-statistic well above 10 indicates that **MOTHERCOLL is a strong instrument for EDUC.**

```
> # Calculate the 95% CI for educ
> confint(mroz2_iv, "educ", level = 0.95)
          2.5 %    97.5 %
educ 0.02751845 0.1481769
```

e.

- Using MOTHERCOLL and FATHERCOLL as instrumental variables, the 95% confidence interval for the coefficient of EDUC is **[0.02751845, 0.1481769]**.
- The interval is **narrower** than the one obtained in part c. This demonstrates that adding another valid instrument variable improves the efficiency of the 2SLS estimator.

f.

- F-statistic = **56.963**
- P-value = **2.2e-16**
- The highly significant p-value indicates that parents' college education jointly explains the

variation in the wife's education. Since the F-statistic remains substantially larger than 10, we conclude that MOTHERCOLL and FATHERCOLL together constitute an **adequately strong** set of instruments.

9.

- To test the validity of the surplus instrument (overidentifying restrictions test), we refer to the Sargan test statistic from the diagnostic tests.
- The Sargan test statistic is **0.238** with a corresponding p-value of **0.626**.
- The null hypothesis of the Sargan test is that all instruments are uncorrelated with the structural error term. Since the p-value is much larger than any conventional significance level (e.g. 0.05 or 0.1), we **fail to reject the null hypothesis**. We conclude that **our instrumental variables (MOTHERCOLL and FATHERCOLL) are jointly valid and exogenous**, supporting the reliability of our 2SLS estimates.

Diagnostic tests:

	df1	df2	statistic	p-value
Weak instruments	2	423	56.963	<2e-16 ***
Wu-Hausman	1	423	0.519	0.472
Sargan	1	NA	0.238	0.626

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.6679 on 424 degrees of freedom

Multiple R-Squared: 0.153, Adjusted R-squared: 0.147

Wald test: 9.724 on 3 and 424 DF, p-value: 3.224e-06

10.20 The CAPM [see Exercises 10.14 and 2.16] says that the risk premium on security j is related to the risk premium on the market portfolio. That is

$$r_j - r_f = \alpha_j + \beta_j(r_m - r_f)$$

where r_j and r_f are the returns to security j and the risk-free rate, respectively, r_m is the return on the market portfolio, and β_j is the j th security's "beta" value. We measure the market portfolio using the Standard & Poor's value weighted index, and the risk-free rate by the 30-day LIBOR monthly rate of return. As noted in Exercise 10.14, if the market return is measured with error, then we face an errors-in-variables, or measurement error, problem.

- Use the observations on Microsoft in the data file `capm5` to estimate the CAPM model using OLS. How would you classify the Microsoft stock over this period? Risky or relatively safe, relative to the market portfolio?
- It has been suggested that it is possible to construct an IV by ranking the values of the explanatory variable and using the rank as the IV, that is, we sort $(r_m - r_f)$ from smallest to largest, and assign the values $RANK = 1, 2, \dots, 180$. Does this variable potentially satisfy the conditions IV1-IV3? Create $RANK$ and obtain the first-stage regression results. Is the coefficient of $RANK$ very significant? What is the R^2 of the first-stage regression? Can $RANK$ be regarded as a strong IV?
- Compute the first-stage residuals, $\hat{v}$, and add them to the CAPM model. Estimate the resulting augmented equation by OLS and test the significance of $\hat{v}$ at the 1% level of significance. Can we conclude that the market return is exogenous?
- Use $RANK$ as an IV and estimate the CAPM model by IV/2SLS. Compare this IV estimate to the OLS estimate in part (a). Does the IV estimate agree with your expectations?
- Create a new variable $POS = 1$ if the market return $(r_m - r_f)$ is positive, and zero otherwise. Obtain the first-stage regression results using both $RANK$ and POS as instrumental variables. Test the joint significance of the IV. Can we conclude that we have adequately strong IV? What is the R^2 of the first-stage regression?
- Carry out the Hausman test for endogeneity using the residuals from the first-stage equation in (e). Can we conclude that the market return is exogenous at the 1% level of significance?
- Obtain the IV/2SLS estimates of the CAPM model using $RANK$ and POS as instrumental variables. Compare this IV estimate to the OLS estimate in part (a). Does the IV estimate agree with your expectations?
- Obtain the IV/2SLS residuals from part (g) and use them (not an automatic command) to carry out a Sargan test for the validity of the surplus IV at the 5% level of significance.

a.

$$\text{Intercept} = 0.003250$$

$$x - mkt = 1.201840 \quad p > 1 \Rightarrow \text{波動比市場大}$$

b.

$$IV1 \text{ (} z \text{ doesn't affect } y \text{)}$$

$$IV2 \text{ (exogenous):}$$

$$IV3 \text{ (relevance): } t = 43.1, p\text{-value} \approx 0$$

$$p\text{-value} = < 2e-16 \text{ significant}$$

$$R^2 = 0.9126$$

$$F = 1858 \text{ (一般看是否} > 10 \text{)}$$

c.

$$\hat{v} \Rightarrow p\text{-value} = 0.0428 \text{ exogenous}$$

d.

$$x - mkt = 1.298318$$

★自變數的衡量誤差會導致OLS算出來的斜率「被稀釋」，變得更平。

e.

$$F = 951.3, p\text{-value} = < 2.2e-16 \approx 0 \text{ significant}$$

$$R^2 = 0.9149$$

f.

$$p\text{-value} = 0.0287 \quad p > 0.01 \Rightarrow \text{won't reject } H_0$$

g.

$$x - mkt = 1.283118 \text{ 比(a)大}$$

h.

$$\text{Sargan-statistic} = 0.5584634$$

Fail to reject H_0 ☹

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.003250	0.006036	0.538	0.591
x_mkt	1.201840	0.122152	9.839	<2e-16 ***

 Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.08083 on 178 degrees of freedom
 Multiple R-squared: 0.3523, Adjusted R-squared: 0.3486
 F-statistic: 96.8 on 1 and 178 DF, p-value: < 2.2e-16

10.20

a.

- The OLS estimate for the coefficient of the market return (beta) is **1.201840**.
- Since the estimated beta is greater than 1, we classify the Microsoft stock as **risky** relative to the market portfolio over this period. It indicates that the stock's returns are more volatile than the overall market.

b.

- The first-stage regression shows that the coefficient of RANK is very significant (p-value < 2e-16).
- The R-squared of the first-stage regression is **0.9126**.
- The F-statistic is **1858**.
- RANK can be regarded as a **very strong IV**. The F-statistic is greater than the rule-of-thumb threshold of 10. It also satisfies the IV conditions: it is highly correlated with the explanatory variable (relevance, IV3), and by using rank instead of actual returns, it potentially filters out extreme measurement errors, acting as an exogenous variable (IV1, IV2).

c.

- Adding the first-stage residuals ($\hat{v}$) to the CAPM model, the p-value for the significance of $\hat{v}$ is **0.0428**.
- At the 1% level of significance, since $0.0428 > 0.01$, we **fail to reject the null hypothesis**. Therefore, we can conclude that the market return is exogenous at the 1% significance level.

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.003018	0.006044	0.499	0.618
x_mkt	1.278318	0.128011	9.986	<2e-16 ***

d.

- Using RANK as an IV, the 2SLS estimate for the market return coefficient is **1.278318**.
- This IV estimate perfectly **agrees with our expectations**. It is **larger than the OLS estimate in part (a)**. This is because measurement error in the independent variables causes **"attenuation bias"** (shrinking the OLS estimate toward zero). **The IV method corrects this bias, resulting in a larger beta.**

e.

- Using RANK and POS as instruments, the F-statistic for testing their joint significance is **951.3**, with a p-value of **<2.2e-16**.
- The R-squared of this first-stage regression is **0.9149**.
- Since the p-value is extremely small, the instruments are jointly significant. With an F-statistic of 951.3 (much higher than 10), we conclude that we have an **adequately strong** set of IVs.

f.

- Carrying out the Hausman test using the residuals from part (e), the p-value is **0.0287**.
- At the 1% level of significance, $0.0287 > 0.01$. We **fail to reject the null hypothesis**. Thus, we conclude that the market return is **exogenous** at the 1% level of significance.

g.

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.003004	0.005972	0.503	0.6157
x_mkt	1.283118	0.126344	10.156	<2e-16 ***
v_hat_2	-0.954918	0.433062	-2.205	0.0287 *

- The IV/2SLS estimate using both RANK and POS is **1.283118**.
- The IV estimate **agrees with our expectations**. Similar to part (d), **it is larger than the original OLS estimate in part (a)**, demonstrating that the multiple instrumental variables successfully corrected the attenuation bias caused by measurement error.

h.

- Carrying out the Sargan test manually for the validity of the surplus IV, the calculated Sargan test statistic is **0.5584634**.
- At the 5% level of significance, the critical value for a Chi-square distribution with 1 degree of freedom is 3.841. Since our statistic is much smaller than the critical value, we **fail to reject the null hypothesis**. We conclude that the surplus instrument is valid, meaning **our instrumental variables are clean and exogenous**.

```
> print(sargan_statistic)
[1] 0.5584634
>
> # Chi-Square test
> #df = L-B = 1
> # Calculate the critical value
> critical_value <- qchisq(0.95, df = 1)
>
> if (sargan_statistic > critical_value){
+   cat("Reject H0: at least one of the IVs is correlated with error")
+ } else{
+   cat("Fail to reject H0: The two IVs are valid!")
+ }
Fail to reject H0: The two IVs are valid!
```